

The Greek Compass: Steering Through Derivatives Risk

A Practitioner's Guide to Mastering Derivatives Risk with the Greeks

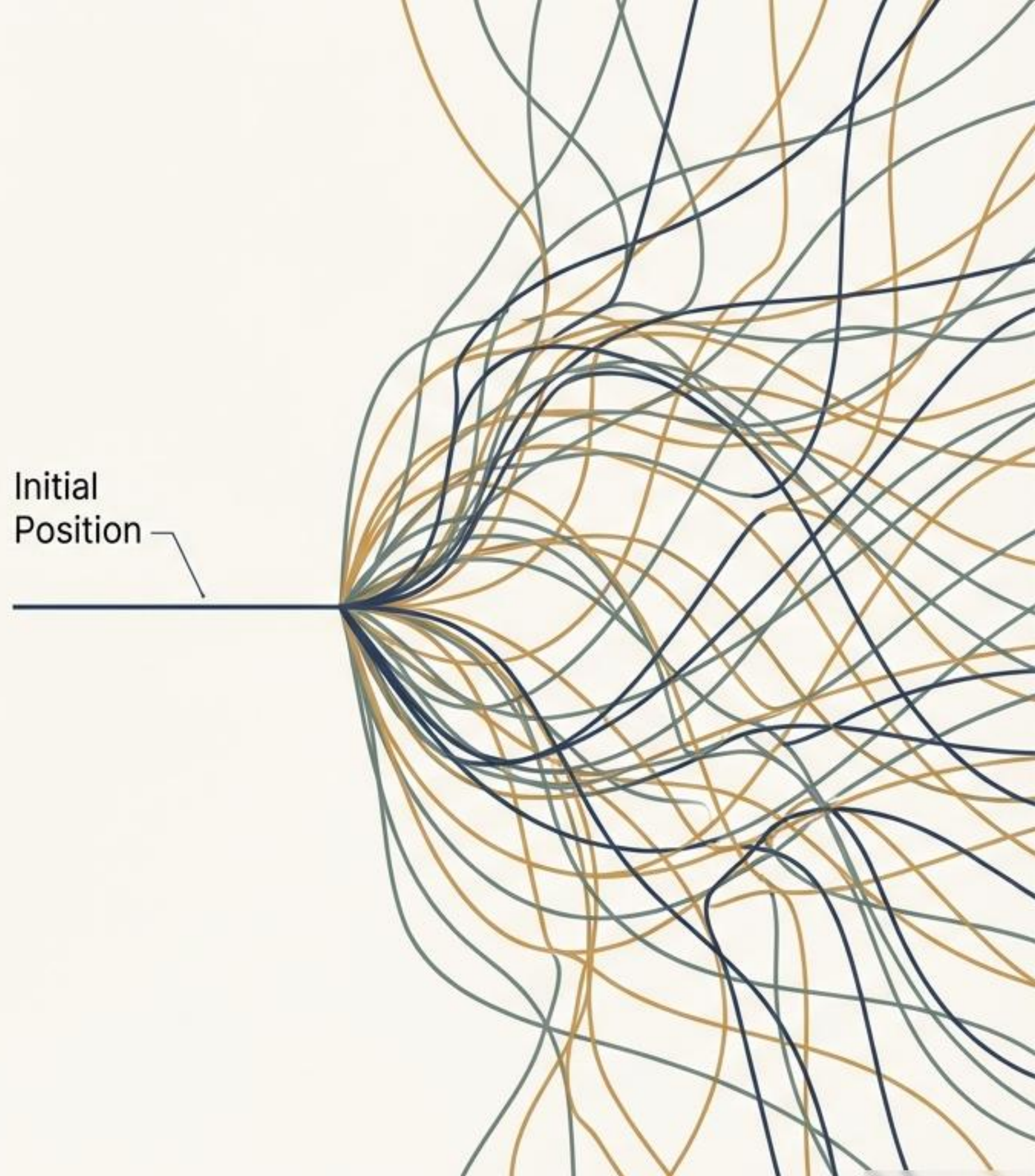
By Michael Fernandes



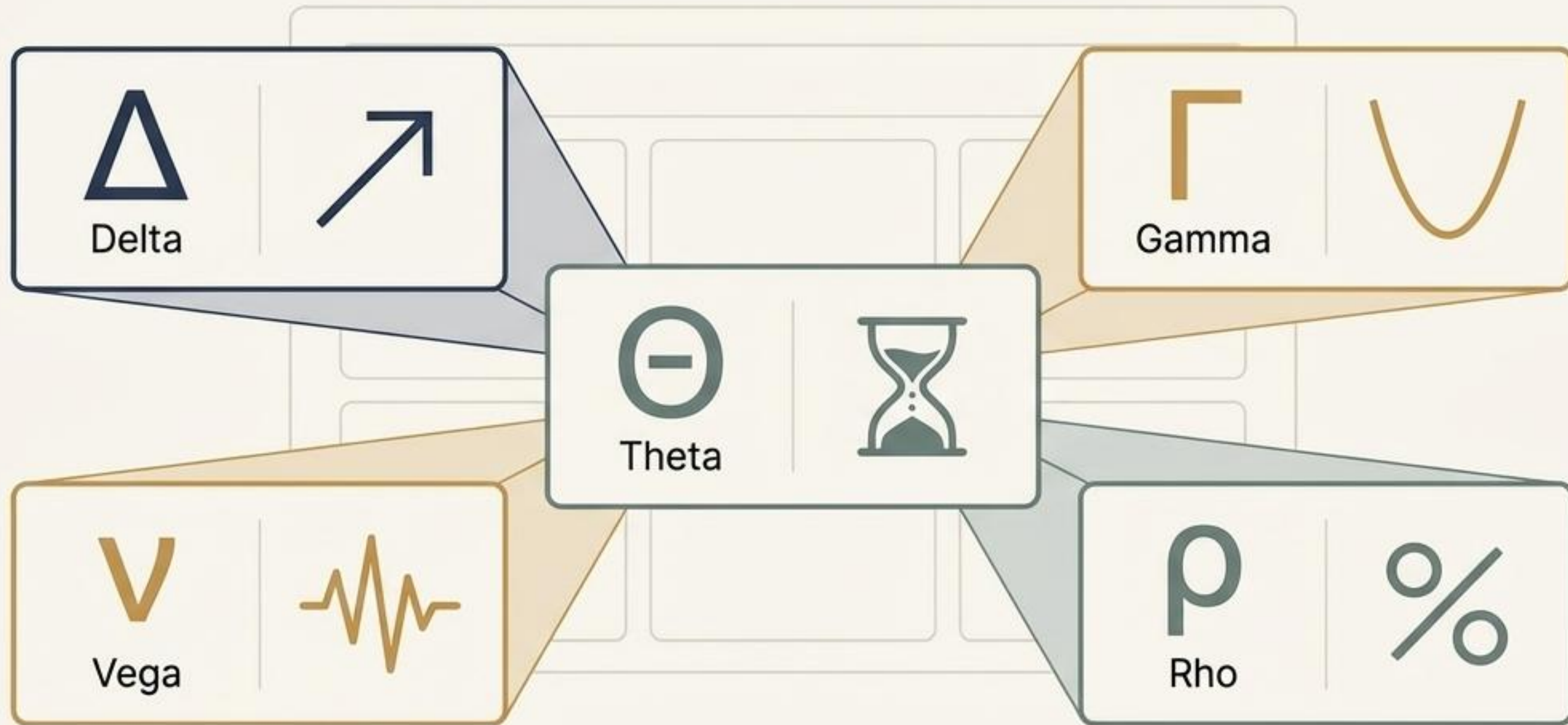
Navigating the Challenge of Non-Linear Risk

- Derivatives, particularly options, possess complex risk profiles that cannot be captured by a single sensitivity measure. Their value changes non-linearly with respect to the underlying asset price, time, volatility, and interest rates.
- Managing a derivatives portfolio without a quantitative framework is akin to flying a high-performance aircraft without instruments—exposing the portfolio to unquantified and potentially catastrophic risks.
- This presentation introduces the framework used by professionals to dissect, measure, and manage these multi-dimensional exposures.

Initial
Position



The Greeks: The Instruments for Precision Risk Management



Core Concept: The Greeks are a set of fundamental risk metrics that translate the abstract concept of an option's value sensitivity into measurable, hedgeable exposures.

Mathematical Foundation: Each Greek represents a partial derivative of an option pricing model (e.g., Black-Scholes) with respect to one of its underlying variables. They provide the quantitative framework that underpins modern derivatives risk management.

Key Property: Portfolio Greeks are additive. The total exposure of a portfolio is the simple sum of the Greeks of its individual positions, allowing for a consolidated view of risk.



Δ Delta: The Speedometer for Directional Exposure

The Concept

Measures the directional sensitivity of an option's price to a change in the price of the underlying asset. It is a first-order derivative: $\Delta = \partial V / \partial S$.

The Measurement

- For every £1 increase in the underlying, an option's value will change by approximately the value of its Delta.
- Example: A call option with a Delta of 0.75 will increase by £0.75 for a £1 rise in the underlying.
- Range: 0 to 1 for calls; -1 to 0 for puts.

The Insight

Delta serves a critical dual purpose for practitioners:

1. **Hedge Ratio:** The number of shares of the underlying asset needed to create a directionally neutral (delta-hedged) position.
2. **Probability Proxy:** Approximates the probability of the option expiring in-the-money.

Gamma: The Accelerometer for Convexity Risk

The Concept:

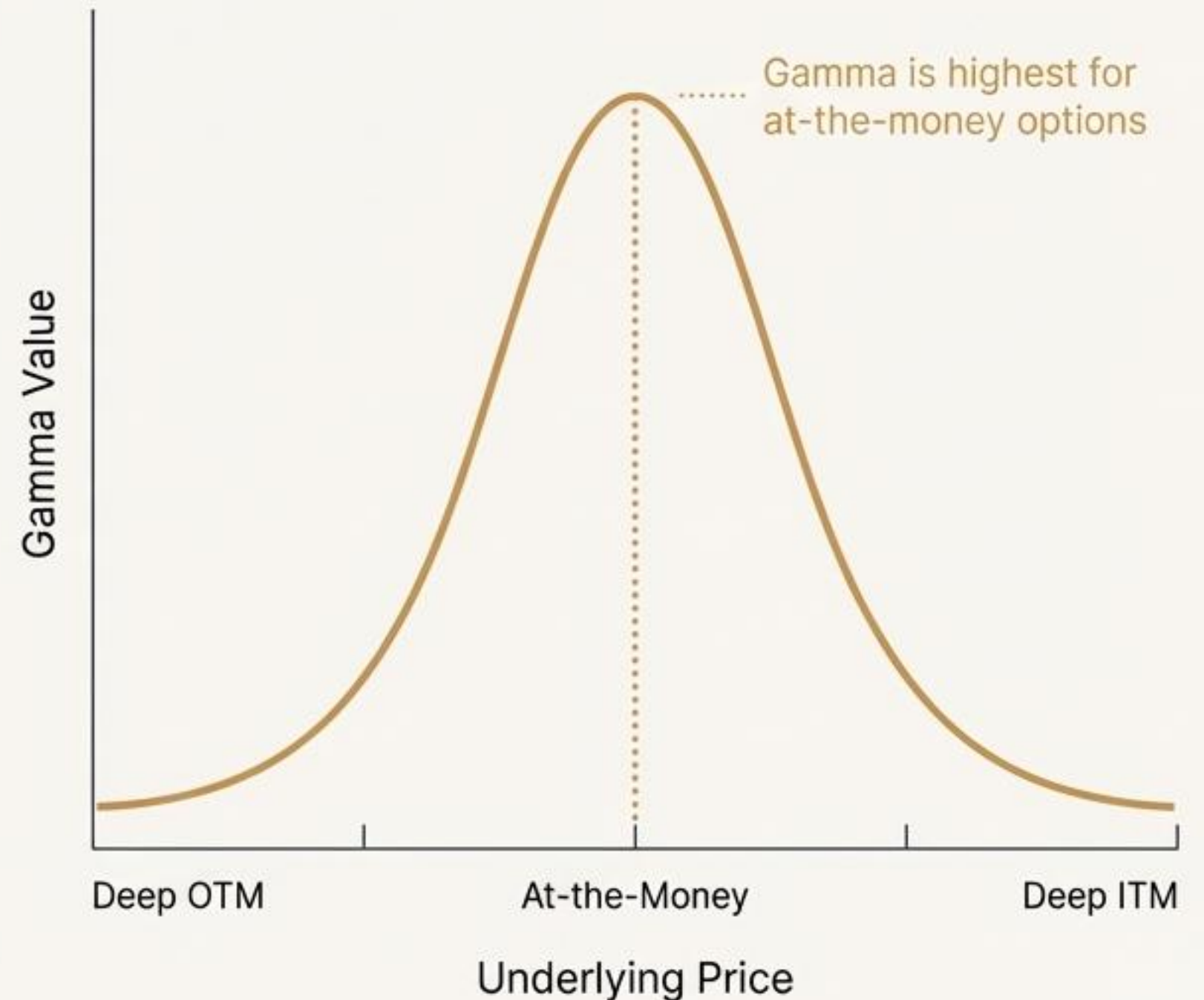
Measures the rate of change of Delta itself for a £1 move in the underlying asset. It is a second-order derivative: $\Gamma = \partial^2 V / \partial S^2$.

The Measurement:

- Example: If an option has a Delta of 0.60 and a Gamma of 0.10, a £1 increase in the underlying will cause its Delta to increase to 0.70.
- Gamma is always positive for long options positions.

The Insight:

Gamma quantifies the error in a static delta hedge. High gamma means Delta is highly unstable, requiring more frequent and costly re-hedging as the underlying price moves. This is the source of 'hedging slippage'.





Theta: The Fuel Gauge for Time Decay

The Concept:

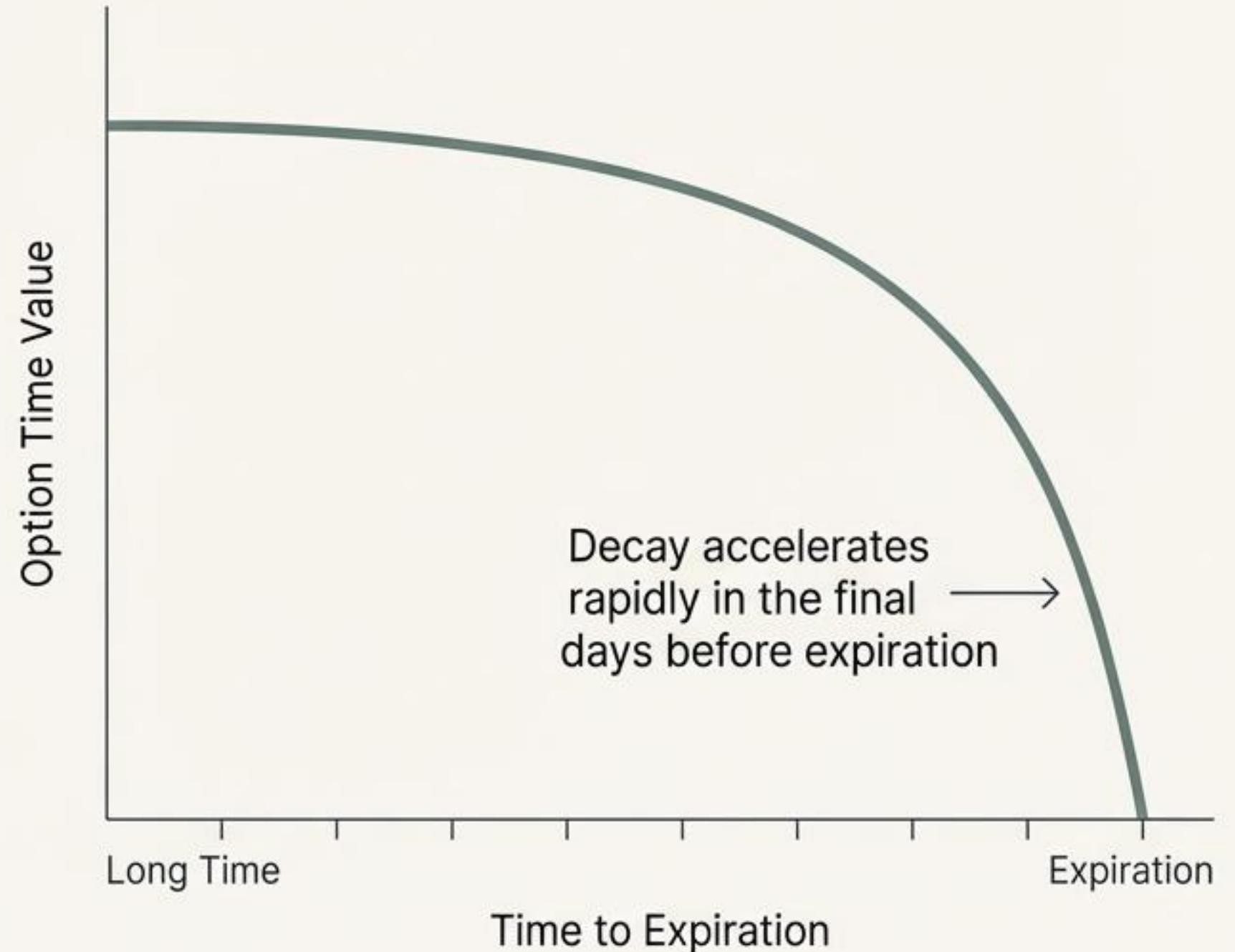
Measures the rate at which an option's value erodes with the passage of time, holding all other factors constant. It is defined as $\Theta = \partial V / \partial \tau$.

The Measurement:

- Typically expressed as a negative value, representing the amount of value lost per day.
- Example: A Theta of -0.25 means the option loses £0.25 in value each day due to time decay.

The Insight:

The rate of time decay is not linear; it accelerates as an option approaches its expiration date. This effect is most pronounced for at-the-money options, which have the most uncertainty to lose.

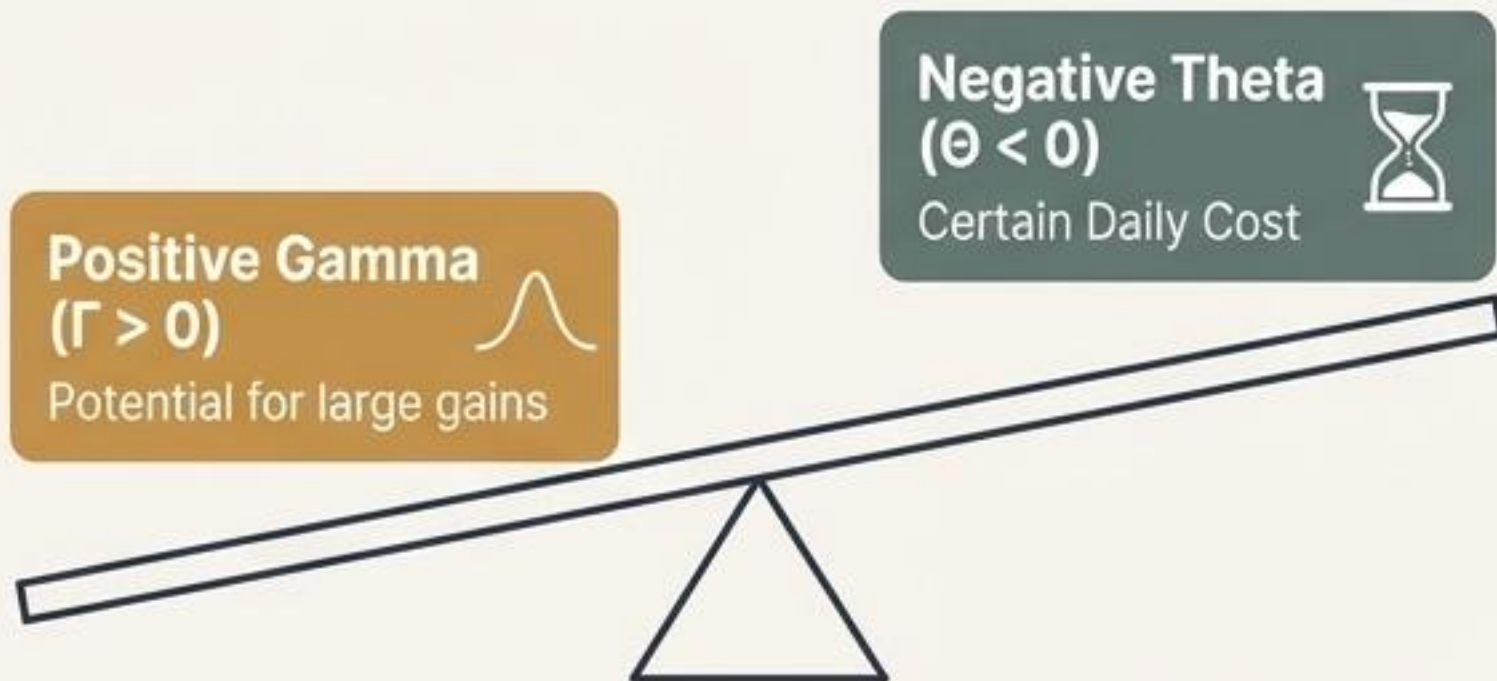


The Practitioner's Dilemma: The Gamma-Theta Trade-Off

For any options position, there is an intrinsic and inescapable trade-off between Gamma and Theta. One cannot be managed without affecting the other.

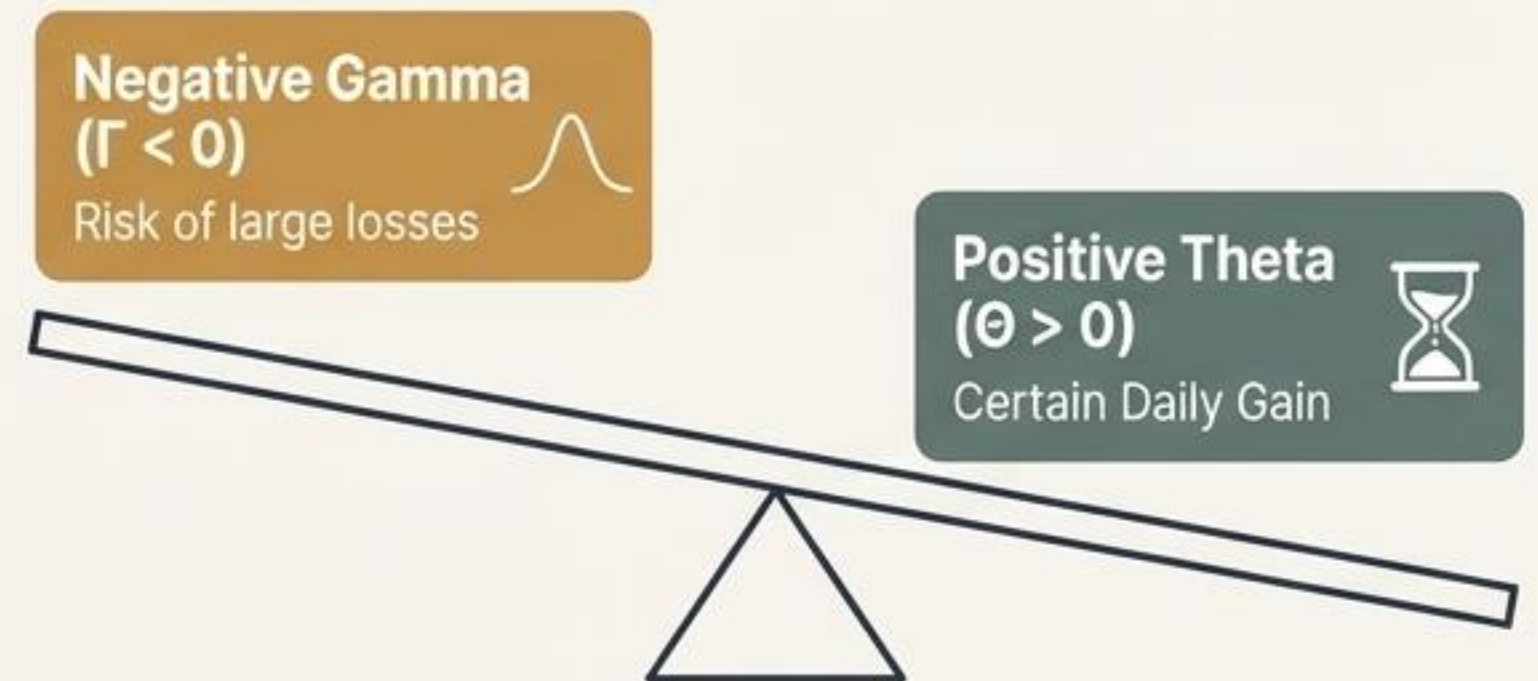
Scenario 1: Long Options (Buying Convexity)

- **Exposure:** Positive Gamma ($\Gamma > 0$), Negative Theta ($\Theta < 0$).
- **Profile:** You benefit from large price movements (convexity) but pay a constant 'premium' through time decay. You are 'paying for the right to be wrong'.



Scenario 2: Short Options (Selling Convexity)

- **Exposure:** Negative Gamma ($\Gamma < 0$), Positive Theta ($\Theta > 0$).
- **Profile:** You collect a steady premium from time decay but face unlimited risk from large price movements. You are 'being paid to be right'.





Vega: The Volatility Gauge

The Concept

Measures the sensitivity of an option's price to a 1% change in the implied volatility of the underlying asset. It is defined as $v = \partial V / \partial \sigma$.

The Measurement

- Vega is always positive for both long calls and puts. An increase in implied volatility increases the value of long options.
- Vega is identical for calls and puts with the same strike and expiry.

The Insight

Vega is highest for at-the-money options with longer terms to expiration. This makes long-dated options powerful tools for expressing a view on future volatility, separate from a view on direction.

Managing vega is as critical as managing delta for a sophisticated options book.



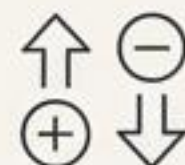
Rho: The Interest Rate Gauge

The Concept



Measures the sensitivity of an option's price to a 1% change in the risk-free interest rate.

The Measurement



- Positive for calls: Higher interest rates increase the value of calls (due to the lower present value of the strike price).
- Negative for puts: Higher interest rates decrease the value of puts.

The Insight



For short-dated options, Rho's impact is often negligible and disregarded in day-to-day management. However, its importance becomes significant for long-dated instruments like LEAPS (Long-Term Equity Anticipation Securities), where the effect of discounting the strike price over many years is substantial.

The Dashboard at a Glance: A Reference Guide

Greek (Symbol)			Measures Sensitivity To	Key Behaviour	Practitioner's Note
	Delta		Underlying Price	Approaches 1 for deep ITM calls, -1 for deep ITM puts	Dual use as hedge ratio & probability proxy
	Gamma		Rate of Change of Delta	Peaks for at-the-money (ATM) options	Quantifies hedging error and cost of rebalancing
	Theta		Time Passage	Decay accelerates near expiration, especially for ATM options	The unavoidable cost of being long options; the income from being short
	Vega		Implied Volatility	Peaks for ATM options; increases with time to expiry	Separates volatility risk from directional risk
	Rho		Interest Rates	Effect is most pronounced for long-dated, ITM options	Often negligible for short-dated options; critical for LEAPS

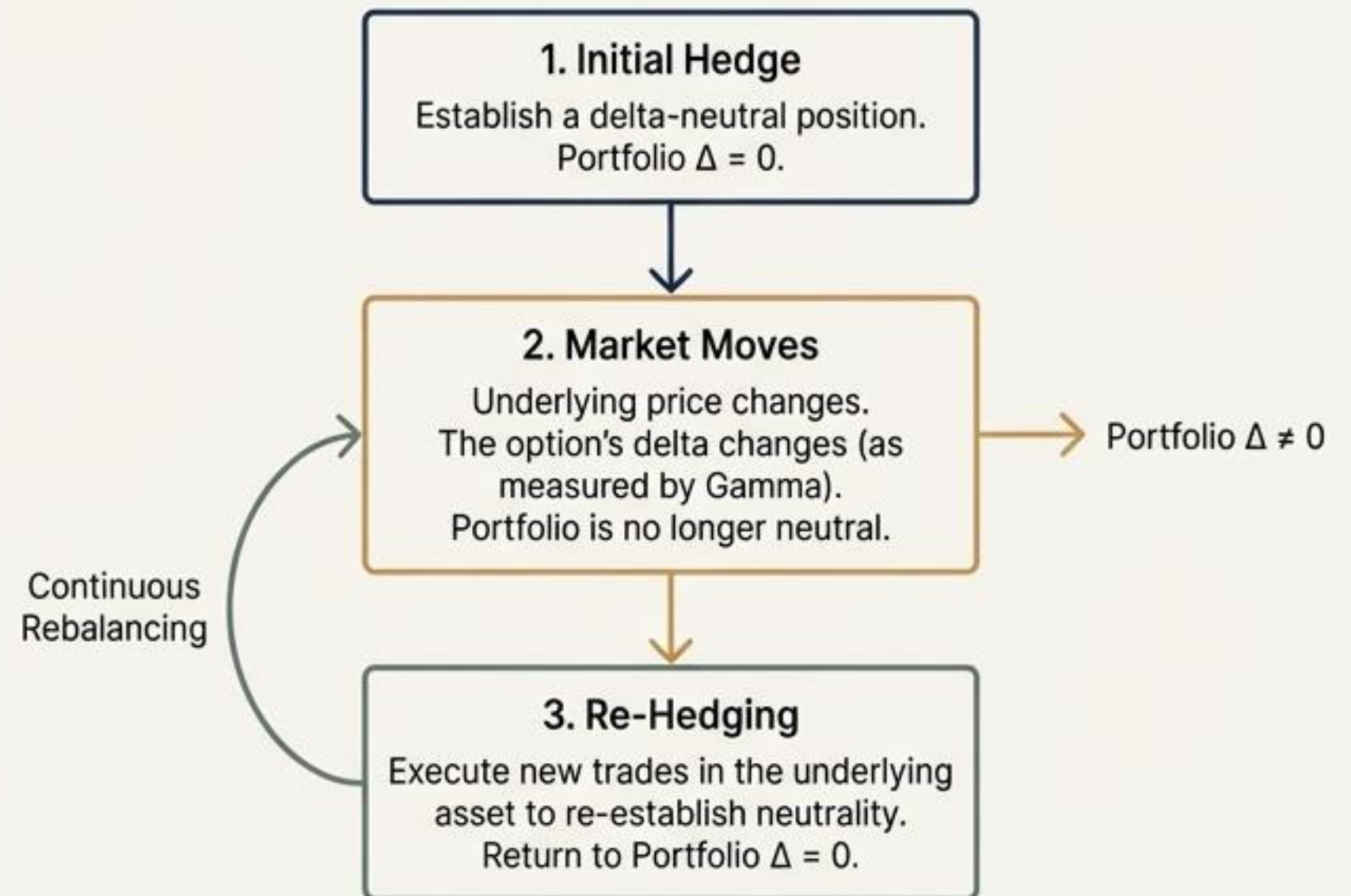
Application I: Dynamic Hedging and Position Monitoring

Position Monitoring



- Traders and risk managers calculate portfolio-level Greeks in real-time to maintain a constant understanding of their net exposures.
- Risk limits are often set in terms of Greeks (e.g., maximum delta, gamma, or vega exposure per desk) to prevent the build-up of concentrated risks.

Dynamic Hedging in Action



Application II: Decomposing Profit and Loss

Core Idea : A portfolio's daily P&L is not a monolith. It can be broken down to attribute gains and losses to the specific risk factors the portfolio was exposed to.

Total P&L	$\approx$	Δ	Delta P&L: From directional movements in the underlying.
		Γ	Gamma P&L: From the convexity effect (realised vs implied volatility).
		Θ	Theta P&L: From the daily time decay.
		ν	Vega P&L: From changes in the level of implied volatility.
		ρ	Rho P&L: From shifts in interest rates.

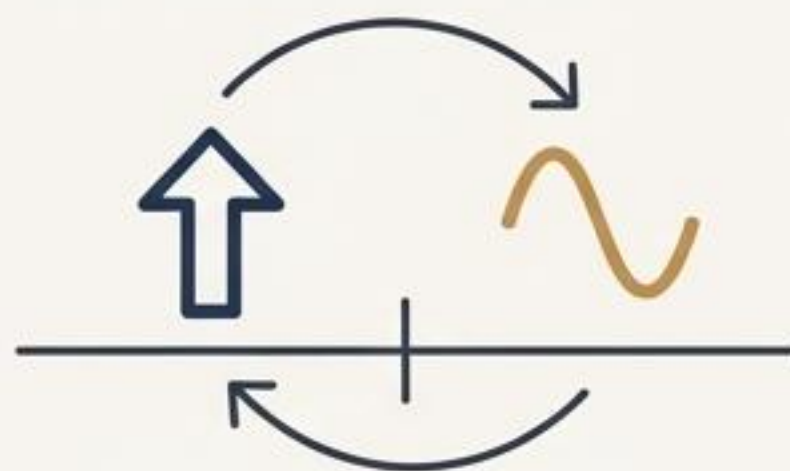
The Insight: This decomposition provides a powerful feedback loop, revealing whether returns were generated from the intended strategy (e.g., a successful or from unintended exposures (e.g., an unhedged directional move).

Beyond the Core Dashboard: An Introduction to Second-Order Greeks

****Concept**:** For highly sophisticated risk management, particularly in volatility trading, **second-order Greeks** provide a **deeper layer of insight**. They measure how the primary Greeks change in relation to other factors.

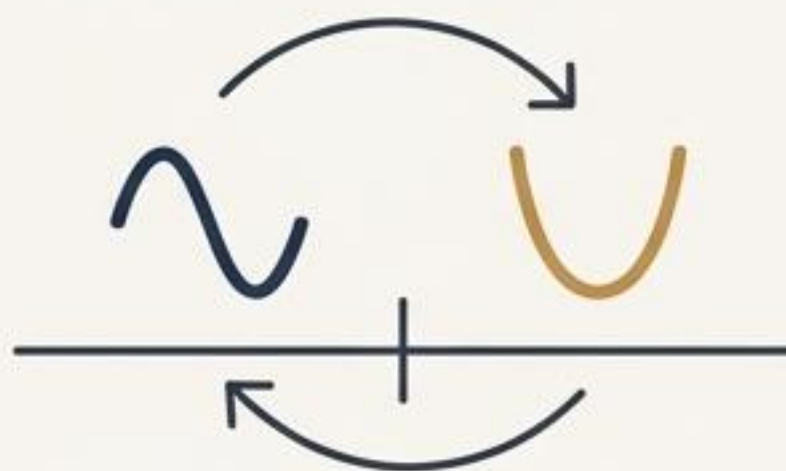
Vanna ($\partial^2 V / \partial S \partial \sigma$)

Measures how Delta changes with a shift in volatility. Essential for managing delta-hedges on a volatility-focused portfolio.



Volga ($\partial^2 V / \partial \sigma^2$)

Measures the convexity of Vega. It is the “gamma of vega,” showing how vega changes as volatility itself changes.



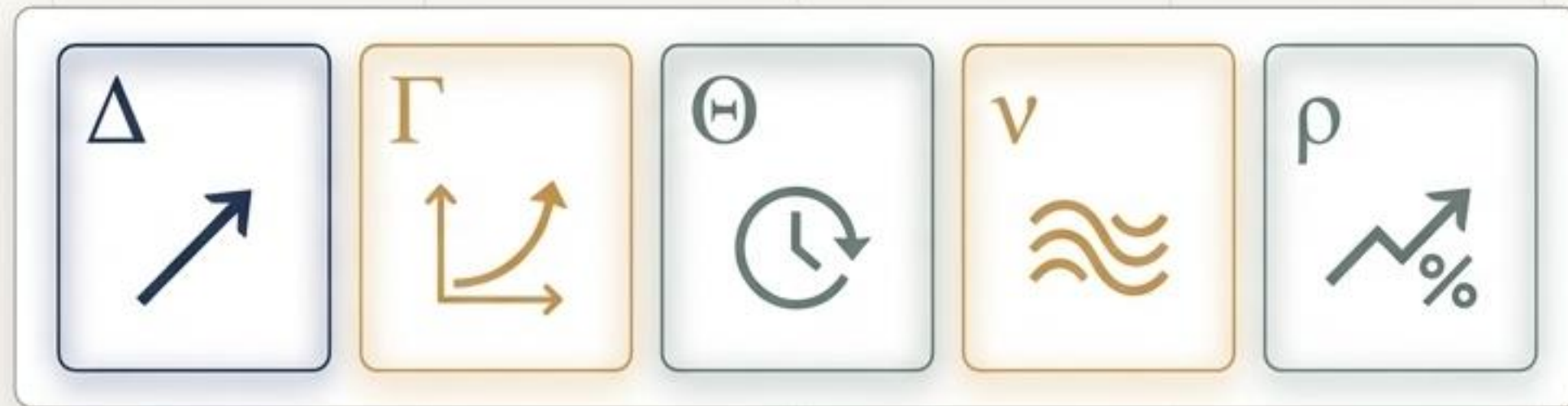
Charm ($\partial^2 V / \partial S \partial \tau$)

Also known as “Delta Decay.” Measures how Delta changes with the passage of time. Critically important for options nearing expiration.



The Complete View: From Blindness to Insight

The Greeks are more than a collection of sensitivities; they are an integrated system for viewing and managing risk.

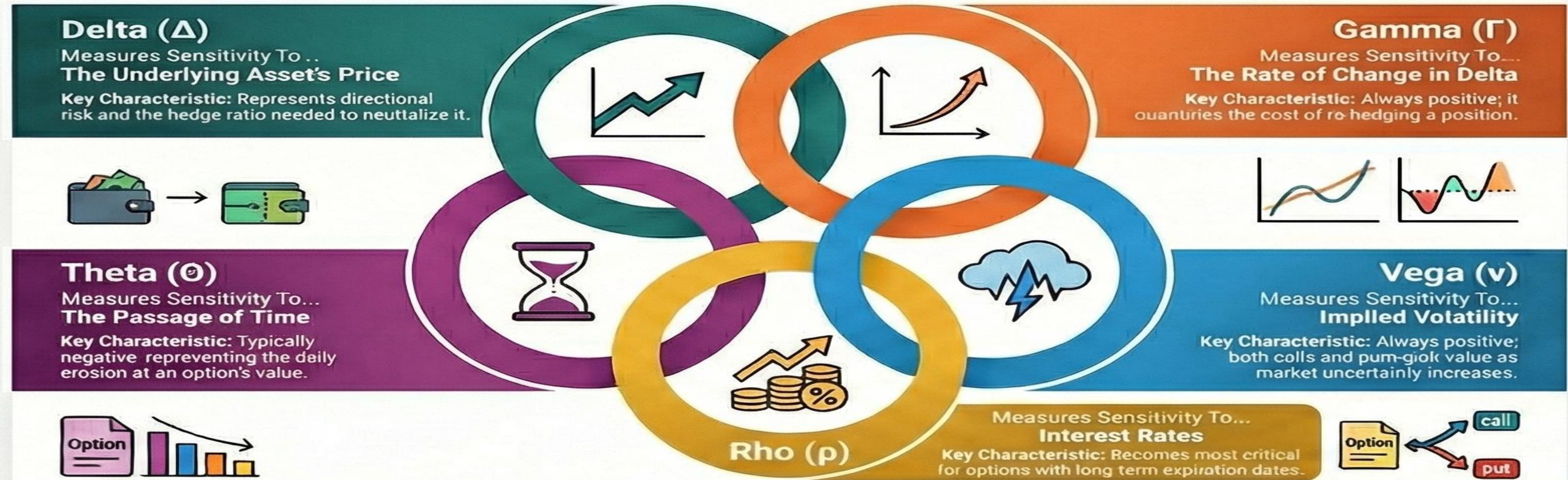


- **Delta** is your direction and speed.
- **Gamma** is your acceleration and instability.
- **Theta** is your fuel consumption over time.
- **Vega** is your sensitivity to turbulence (volatility).

The Key Takeaway: By assembling this dashboard, practitioners transform the complex, abstract nature of derivatives risk into a set of precise, measurable, and manageable exposures. This quantitative framework is the indispensable foundation of modern derivatives trading and risk management.

Decoding the Greeks: A Guide to Options Risk

The "Greeks" are essential risk metrics used in derivatives trading, representing how sensitive an option's price is to different factors for hedging and managing portfolio risk.



How Traders Use the Greeks



Position Monitoring

Traders use Greek-based limits to monitor exposures and prevent concentrated risks.



Dynamic Hedging

Positions are continuously rebalanced by trading assets as Greek values change.



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Daily profit and loss is broken down to see which risk factors drove returns.